

Lecture #5: Discrete Probability Distributions

MATH 145 – Calculus I

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2025-10-01

Motivating Example

Example: I offer a lottery game for \$5 a play. I spin a spinner with 1-10 on it, evenly spread out. If it lands on 1-9, I keep your money. If it lands on 10, you win \$100. Let P be the payout you receive from this game.

What is the probability mass function of the random variable P ?

What is the expected value of the random variable P , $E(P)$?

There are many situations like this, where we have a random experiment with two outcomes, a “success” and a “failure”, with a certain probability of success. Is there a way we can take advantage of the similarities?

Bernoulli Distribution

These kinds of random variables can be described using a Bernoulli distribution.

A **Bernoulli random variable** is a random variable with two outcomes, a “*success*”, 1; or a “*failure*”, 0, with one parameter: **success probability** p .

We write this as:

$$X \sim \text{Bernoulli}(p)$$

What does the probability mass function of a Bernoulli(p) look like?

PMF of the Bernoulli Distribution

The probability mass function of the Bernoulli distribution with success probability p is:

$$p_X(x) = P(X = x) = \begin{cases} p & \text{for } x = 1 \\ 1 - p & \text{for } x = 0 \end{cases}$$

Expected Value and Variance of the Bernoulli Distribution

The expected value of a Bernoulli(p) distribution is:

$$E(X) = p.$$

The variance of a Bernoulli(p) distribution is:

$$V(X) = p(1 - p).$$

Can you prove these?

Example: Bernoulli Distribution

Example: You flip a biased coin, landing on heads 65% of the time. If the coin lands on heads, you make \$5, and if it lands on tails, you get nothing. Let X be the payout of the game.

- 1 What is the distribution of X ?

Example: Bernoulli Distribution

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- 1 What is the distribution of X ?
- 2 What is the expected payout?

Example: Bernoulli Distribution

Example: You flip a biased coin, landing on heads 65% of the time. If the coin lands on heads, you make \$5, and if it lands on tails, you get nothing. Let X be the payout of the game.

- 1 What is the distribution of X ?
- 2 What is the expected payout?
- 3 What is the variance and standard deviation of the payouts?

The Binomial Distribution

Suppose instead of flipping a single coin, we flip it 10 times, and record the number of heads, and call that variable X . How is X distributed?

So instead of being a single Bernoulli experiment, we are performing 10 *independent* Bernoulli experiments, and recording the number of “successes”.

This is the idea behind the **binomial distribution**.

The Binomial Distribution

The **binomial distribution** is defined as the sum of n independent and identically distributed Bernoulli random variables with success probability p .

In formal terms:

$$X_i \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$$

$$Y = \sum_{i=1}^n X_i$$

$$Y \sim \text{Binomial}(n, p)$$

The binomial distribution has 2 parameters: the **number of trials** n and the **success probability** p .

PMF of the Binomial Distribution

The probability mass function of a binomially distributed random variable X is:

$$p_X(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x},$$

for $x \in \{0, 1, 2, \dots, n\}$.

Cumulative Distribution Function

There are times when the probability mass function isn't the most useful way to frame a problem.

The **cumulative distribution function (CDF)** is defined as the probability that the random variable X is less than or equal to x .

For a discrete random variable:

$$P(X \leq x) = \sum_{i=1}^x p_X(x).$$

Binomial Distribution Example

Example:

The probability that a patient recovers from a stomach disease is .8. Suppose 20 people are known to have contracted this disease. What is the probability that:

- ❶ exactly 14 recover?
- ❷ at least 10 recover?
- ❸ at least 14 but not more than 18 recover?
- ❹ at most 16 recover?

Expected Value and Variance of the Binomial Distribution

The expected value of a Binomial(n, p) distribution is:

$$E(X) = np.$$

The variance of a Binomial(n, p) distribution is:

$$V(X) = np(1 - p).$$

Can you prove these?

Binomial Distribution Example

Example:

An oil exploration firm is formed with enough capital to finance ten explorations. The probability of a particular exploration being successful is .1. Assume the explorations are independent. Find the mean and variance of the number of successful explorations.

Suppose the firm has a fixed cost of \$20,000 in preparing equipment prior to doing its first exploration. If each successful exploration costs \$30,000 and each unsuccessful exploration costs \$15,000, find the expected total cost to the firm for its ten explorations.

The Geometric Distribution

Let's consider a new game: We are going to flip the coin until we get a tails. We will pay out \$1 for each coin flip it takes until we get a tails.

- If the first coin flip is a tails (T), you get \$1.

The Geometric Distribution

Let's consider a new game: We are going to flip the coin until we get a tails. We will pay out \$1 for each coin flip it takes until we get a tails.

- If the first coin flip is a tails (T), you get \$1.
- If the first coin flip is a heads and the second coin flip is a tails (HT), you get \$2.

The Geometric Distribution

Let's consider a new game: We are going to flip the coin until we get a tails. We will pay out \$1 for each coin flip it takes until we get a tails.

- If the first coin flip is a tails (T), you get \$1.
- If the first coin flip is a heads and the second coin flip is a tails (HT), you get \$2.
- If the first two coin flips are heads and the third is a tails (HHT), you get \$3.

The Geometric Distribution

Let's consider a new game: We are going to flip the coin until we get a tails. We will pay out \$1 for each coin flip it takes until we get a tails.

- If the first coin flip is a tails (T), you get \$1.
- If the first coin flip is a heads and the second coin flip is a tails (HT), you get \$2.
- If the first two coin flips are heads and the third is a tails (HHT), you get \$3.
- If the first three coin flips are heads and the fourth is a tails ($HHHT$), you get \$4.

The Geometric Distribution

Let's consider a new game: We are going to flip the coin until we get a tails. We will pay out \$1 for each coin flip it takes until we get a tails.

- If the first coin flip is a tails (T), you get \$1.
- If the first coin flip is a heads and the second coin flip is a tails (HT), you get \$2.
- If the first two coin flips are heads and the third is a tails (HHT), you get \$3.
- If the first three coin flips are heads and the fourth is a tails ($HHHT$), you get \$4.
- And so on and so on...

The Geometric Distribution

Let X be the pay out of this game.

- What is the sample space for this game?

The **geometric distribution** is a random variable which counts the number of trials until the first success, if each trial is independent and has the same probability of success p .

The geometric distribution has only one parameter, the **success probability** p .

The Geometric Distribution

Let X be the pay out of this game.

- What is the sample space for this game?
- What might the probability mass function look like?

The **geometric distribution** is a random variable which counts the number of trials until the first success, if each trial is independent and has the same probability of success p .

The geometric distribution has only one parameter, the **success probability** p .

PMF of the Geometric Distribution

The probability mass function of a random variable X with a geometric distribution is:

$$p_X(x) = P(X = x) = (1 - p)^{x-1}p,$$

for $x \in \{1, 2, 3, \dots\}$.

Geometric Distribution Example

Example:

Consider the game from earlier. Suppose the coin lands on tails 70% of the time. What is the probability that:

- ① the game ends on the third coin flip?
- ② the game lasts 2 or fewer rounds?
- ③ the game lasts more than two rounds?
- ④ the game lasts more than four rounds, given that it lasted two rounds already?

Expected Value and Variance of the Geometric Distribution

The expected value of a $\text{Geometric}(p)$ distribution is:

$$E(X) = \frac{1}{p}.$$

The variance of a $\text{Geometric}(p)$ distribution is:

$$V(X) = \frac{1-p}{p^2}.$$

Can you prove these? (Consider a recursive definition. The textbook also uses an argument based on the derivative of the sum of a geometric series.)

Geometric Distribution Example

Example:

Of a population of consumers, 60% are reputed to prefer a particular brand, A , of toothpaste. If a group of randomly selected consumers is interviewed, what is the mean and standard deviation of the number of consumers that must be interviewed before someone who doesn't like the toothpaste is found?

The Negative Binomial Distribution

We can further generalize the idea behind the geometric distribution. What if instead of waiting until the *first* success, we wait until the second success, or the third success, or the r^{th} success?

- What is the sample space for this distribution?

The **negative binomial distribution** is a random variable which counts the number of trials until the r^{th} success, if each trial is independent and has the same probability of success p .

The negative binomial distribution has two parameters, the **number of successes** r and **success probability** p .

The Negative Binomial Distribution

We can further generalize the idea behind the geometric distribution. What if instead of waiting until the *first* success, we wait until the second success, or the third success, or the r^{th} success?

- What is the sample space for this distribution?
- What might the probability mass function look like?

The **negative binomial distribution** is a random variable which counts the number of trials until the r^{th} success, if each trial is independent and has the same probability of success p .

The negative binomial distribution has two parameters, the **number of successes** r and **success probability** p .

PMF of the Negative Binomial Distribution

The probability mass function of a random variable X with a geometric distribution is:

$$p_X(x) = P(X = x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r,$$

for $x \in \{r, r+1, r+2, \dots\}$.

Expected Value and Variance of the Geometric Distribution

The expected value of a $\text{NB}(r, p)$ distribution is:

$$E(X) = \frac{r}{p}.$$

The variance of a $\text{NB}(p)$ distribution is:

$$V(X) = \frac{r(1-p)}{p^2}.$$

Negative Binomial Example

A geological study indicates that an exploratory oil well should strike oil with probability .2.

- ① What is the probability that the first strike comes on the third well drilled?
- ② What is the probability that the third strike comes on the seventh well drilled?
- ③ What assumptions did you make to obtain the answers to parts (1) and (2)?
- ④ Find the mean and variance of the number of wells that must be drilled if the company wants to set up three producing wells.